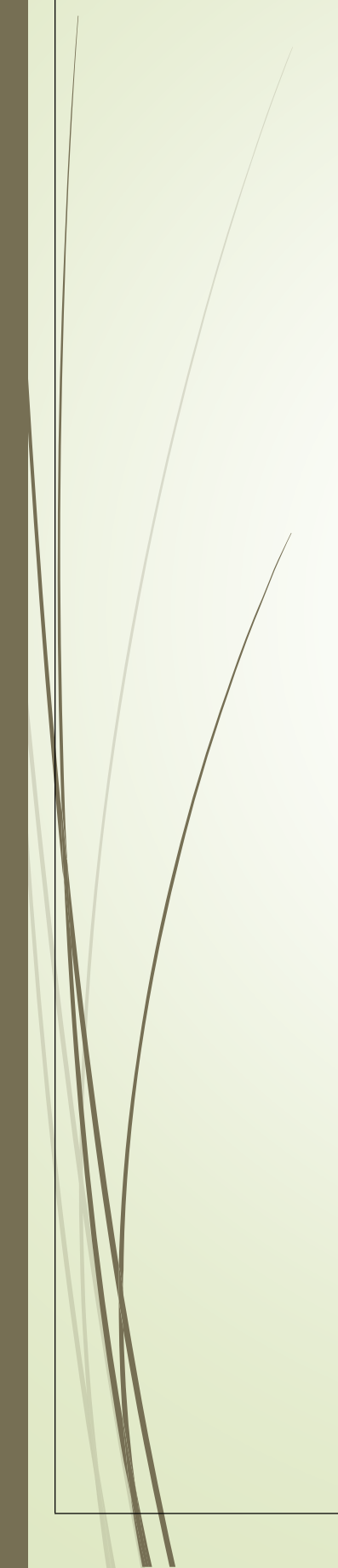


REQUIREMENT ANALYSIS DOCUMENT

RESIN AND RADIANCE

- "Crafting Memories That Shine Forever."
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1.Introduction

Resin and Radiance is a creative business specializing in handmade resin art, customized jewellery, personalized gifts, home dcor items, keychains, bookmarks, coasters, trays, and other resin-based products. Many of the products are handcrafted and made-to-order according to customer preferences such as colours, names, designs, embedded elements, and finishing styles.

2.Purpose

The primary purpose of the **Resin and Radiance Order Management System (OMS)** is to provide a structured database system that supports the complete order lifecycle from customer registration to final product delivery.

The system is designed to:

- Maintain customer information securely.
- Manage product catalogue and inventory.
- Handle customized resin product orders.
- Record and monitor payment transactions.
- Track shipping and delivery status.
- Generate accurate business records.
- Reduce manual errors and duplicate data.
- Improve operational efficiency and customer satisfaction.

3.Major Activities in the System

1. Customer Registration and Login

- Register new customers.
- Authenticate user login.
- Manage customer profiles and addresses.

2. Product Management

- Add new resin products.
- Update product details, prices, and images.
- Remove discontinued products.

3. Inventory Management

- Track product stock levels.
- Update inventory after each order.
- Notify the administrator of low-stock items.

4. Order Management

- Browse and select products.
- Add items to the shopping cart.
- Place orders with customization requests.
- Update order status (Pending, Confirmed, Shipped, Delivered).

5. Custom Order Processing

- Record customer-specific requirements.
- Manage personalized resin designs.
- Track custom order production.

6. Payment Management

- Process customer payments.
- Record transaction details.
- Generate invoices and receipts.
- Update payment status.

7. Shipping and Delivery

- Assign courier services.
- Generate tracking numbers.
- Monitor shipment and delivery status.
- Confirm successful delivery.

8. Customer Communication

- Send order confirmation notifications.
- Notify customers about payment status.
- Provide shipping and delivery updates.

4. Functional Requirements

Functional ID	Functional Requirement	Description
1	Customer Management	Register, login, and manage customer profiles.
2	Product Management	Add, update, view, and delete products.
3	Inventory Management	Track and update product stock.
4	Order Management	Place, process, and track customer orders.
5	Payment Management	Record payments and update payment status.
6	Shipping Management	Manage shipping details and delivery status.
7	Report Generation	Generate sales, inventory, and order reports.
8	Security	Provide secure login and role-based access.

5. Non-Functional Requirements

1. Performance

- The system should process orders quickly.
- Product and customer data should load without delay.

2. Security

- Customer and payment information should be protected.
- Only authorized users should access the system.

3. Reliability

- The system should operate continuously with minimal downtime.
- Data should be stored accurately without loss.

4. Usability

- The system should have a simple and user-friendly interface.
- Users should be able to navigate the system easily.

5. Availability

- The system should be available whenever users need it.

- **Services should remain accessible during business hours.**

6. Scalability

- **The system should support an increasing number of customers, products, and orders.**

7. Maintainability

- **The system should be easy to maintain, update, and fix.**

8. Data Integrity

The system should ensure data accuracy and consistency.

Conclusion

The Resin and Radiance E-Commerce Order Management System provides an efficient and organized solution for managing customers, products, inventory, orders, payments, and shipping. By automating these business processes, the system improves accuracy, reduces manual work, and enhances customer satisfaction. The proposed database system supports smooth day-to-day operations and offers a scalable foundation for the future growth of the business.